

Quantum Well Resonant Tunneling Diode Probe of Correlated States in Twisted Bilayer MoS₂

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Moiré superlattices formed in transition metal dichalcogenides (TMDs) offer a versatile platform for exploring emergent quantum phases arising from strong electronic correlations. In this work, we develop a new experimental platform, the quantum well resonant tunneling diode (QWRTD), to probe the electronic landscape of $\approx 57^\circ$ twisted bilayer MoS₂ (tMoS₂). By measuring the differential conductance (dI/dV_{Probe}) as a function of filling factor ν and displacement field D , we observe a robust integer correlated insulating state at $\nu = 1$ that persists across the measured displacement field range. Furthermore, we identify a displacement-field-induced fractional insulating state at $\nu = 3/4$ for $D < -75$ mV/nm. Temperature- and magnetic-field-dependent measurements characterize this $\nu = 3/4$ state as a potential generalized Wigner crystal. These correlated states are also observed in another device with a similar twist angle ($\approx 56.5^\circ$). Our results provide direct evidence of correlated states in near-AB-stacked tMoS₂ and establish QWRTD as a powerful experimental tool for investigating strong correlations and topology in van der Waals heterostructures.

Introduction.—Moiré superlattices formed by stacking two-dimensional van der Waals materials with a small rotational or lattice mismatch have emerged as an exceptional platform for exploring strongly correlated physics[1–6]. In semiconductor transition metal dichalcogenides (TMDs), the interference pattern between neighboring layers generates a periodic moiré potential that quenches the kinetic energy of charge carriers, confining them into flat mini-bands[7, 8]. To date, most experimental studies on twisted TMD homobilayers and heterobilayers have focused on the hole-doped regime, revealing a rich spectrum of exotic states, including Mott insulators, generalized Wigner crystals, quantum anomalous/spin Hall states, and superconductivity[3, 6, 9–19].

Pioneering research on twisted TMDs focused predominantly on the valence band owing to its simple electronic structure, in which the K and K' valleys' localization driven by in-plane $d_{x^2-y^2}$ and d_{xy} orbitals and giant Ising spin-orbit coupling of hundreds of meV yields a single-band triangular Hubbard model [7]. By contrast, the conduction band exhibits enhanced interlayer hybridization from dominant out-of-plane d_{z^2} orbitals, multi-pocket quantum interference from nearby six-fold Q (Λ) valleys, and significantly weaker spin-orbit coupling. This reduced spin-valley locking renders electron-doped moiré superlattices highly field-tunable, providing a versatile platform for field-induced topological band inversions and multi-flavor electronic phases. Near AB-stacking ($\approx 57^\circ$) twisted bilayer MoS₂ (tMoS₂) offers a promising platform to explore this electron-doped regime, where narrow moiré minibands and strong Coulomb interactions are expected to yield rich correlated electron dy-

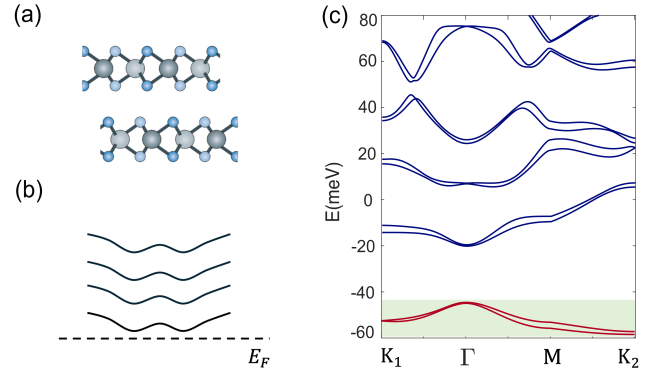


FIG. 1. (a) Schematic picture of the crystal structure of AB-stacked tMoS₂ with 2H-type stacking order. (b) The moiré bands have nearly four-fold degeneracy including spin and valley. (c) Continuum model calculations. The calculated moiré band structure (K valley) of 57° -tMoS₂ at displacement field $D = 0.1$ V/nm.

namics [20]. Unlike AA-stacked TMD bilayers, 2H-type AB stacking breaks in-plane inversion symmetry [Fig. 1(a)], giving rise to distinct layer-polarized moiré potentials. Due to the combined spin and valley degrees of freedom, the low-energy moiré minibands display a nearly four-fold degeneracy [Fig. 1(b)]. Continuum model calculations [see Supplementary Section S1] indicate that the lowest conduction moiré miniband is well isolated from higher-lying bands under a displacement field $D = 0.1$ V/nm [Fig. 1(c)], exhibiting a narrow bandwidth $W \approx 15$ meV at the K valley, weakly split

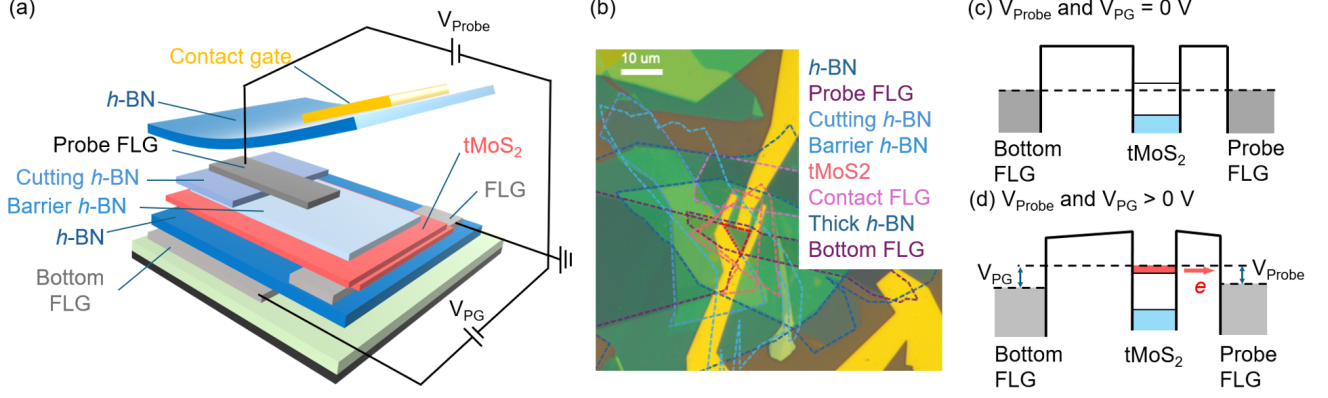


FIG. 2. (a) Schematic structure of a QWRD (contact gate/top *h*-BN/Probe FLG/cutting *h*-BN/barrier *h*-BN/tMoS₂/contact FLG/bottom *h*-BN/bottom FLG). The probe bias V_{Probe} is applied between the Probe FLG and tMoS₂ (via contact FLG), while the probe-gate voltage V_{PG} is applied between the bottom FLG and tMoS₂, together enabling joint control of the tunneling bias and the local displacement field/doping. (b) Optical micrograph of the assembled QWRD device. Distinct constituent layers are outlined by dashed lines of different colors, with the active tunneling region enclosed by the red dashed contour. (c-d) Schematic band alignment of a QWRD when V_{Probe} and $V_{\text{PG}} = 0$ V (c), and V_{Probe} and $V_{\text{PG}} > 0$ V (d).

by spin-orbit coupling. Comparison with the estimated on-site Coulomb interaction $U \approx 25$ meV yields a correlation ratio $U/W \gtrsim 1$, placing the conduction band of nearly AB-stacked tMoS₂ firmly in the strongly correlated regime even under a large displacement field.

However, experimental mapping of this electron-doped flat-band regime remains severely limited. Conventional macroscopic planar transport measurements fail to resolve these flat-band states, hindered by the high Schottky contact barriers at the TMD conduction band edge [6, 21–24] and the inevitable spatial averaging across macroscopic channels due to non-uniform distributions of twist angle and electric displacement field D in large-scale devices [25–28]. While local scanning probes, such as scanning tunneling spectroscopy (STS) and scanning single-electron transistors (SET), effectively mitigate spatial inhomogeneity by isolating local domains [11, 29–31], their open-geometry configuration intrinsically precludes the top gate integration required for independent, quantitative displacement-field (D) control. A technique capable of unifying local spatial resolution with full dual-gate field tunability and efficient electron injection has thus remained elusive.

In this Letter, we introduce a quantum well resonant tunneling diode (QWRD) architecture that enables contact-free, localized tunneling spectroscopy to study strongly correlated, near AB-stacking ($\approx 57^\circ$) tMoS₂ moiré systems. By directly sensing correlated states via vertical tunneling, this platform bypasses in-plane Ohmic contacts while maintaining effective and uniform dual-gate electrostatics. In the electron-doped regime at zero magnetic field, we observe a robust Mott insulating state at integer filling $\nu = 1$ and an emergent

displacement-field-induced fractional insulating state at $\nu = 3/4$. Temperature- and magnetic-field-dependent spectroscopy further identifies this fractional state as a generalized Wigner crystal driven by long-range Coulomb interactions, establishing the QWRD as a powerful probe for multi-flavor moiré physics.

Device architecture and electrostatic mapping.—The QWRD architecture and operational mechanism are illustrated in Fig. 2. The heterostructure comprises an 8-layer vertical stack (top *h*-BN/Probe few-layer graphene (FLG)/cutting *h*-BN/barrier *h*-BN/tMoS₂/contact FLG/bottom *h*-BN/bottom FLG) [Fig. 2(a)], where the key to non-invasive local probing lies in eliminating parasitic transport pathways. Specifically, a patterned cutting *h*-BN layer spatially isolates the active tunneling region from edge leakage, ensuring that the measured signal originates purely from the pristine moiré channel [see Supplementary Section S2]. Fig. 2(b) shows the optical micrograph of the assembled functional QWRD, where constituent 2D layers are outlined by dashed contours. Consistent with our design [Fig. 2(a)], the active tunneling region is spatially confined by the patterned cutting *h*-BN layer, highlighted by the red dashed boundary in Fig. 2(b). As depicted in the schematic band alignments [Figs. 2(c) and 2(d)], applying a probe bias $V_{\text{Probe}} > 0$ V shifts the probe Fermi level relative to the tMoS₂ conduction band, driving a vertical tunneling current I from tMoS₂ to probe-FLG through the thin barrier *h*-BN layer. Concurrently, the voltage V_{PG} , applied to the bottom FLG relative to tMoS₂, capacitively couples to the tMoS₂ channel in conjunction with V_{Probe} : their sum (weighted by the respective gate capacitances) sets the local carrier density n , while their

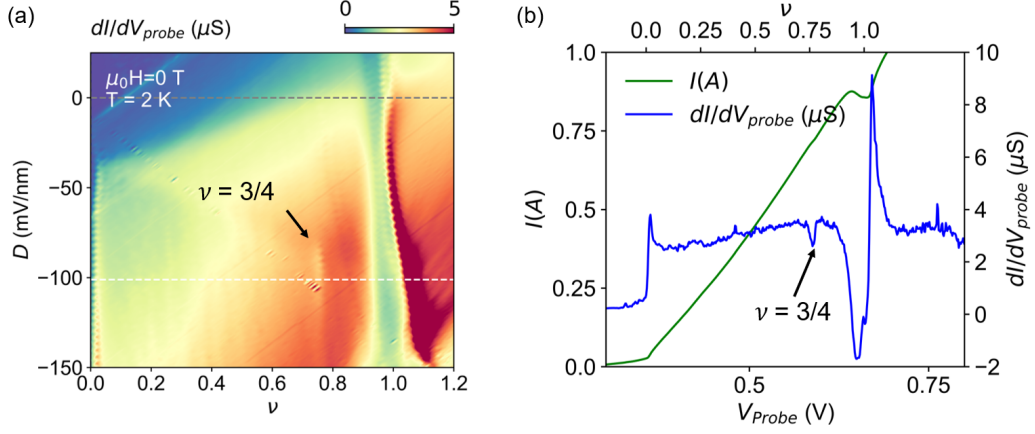


FIG. 3. (a) Measured differential conductance dI/dV_{Probe} map as a function of filling factor ν and vertical electric displacement field D at $T = 2$ K and zero magnetic field. A persistent integer insulating state is present at $\nu = 1$ across all D , while a displacement-field-induced fractional insulating state at $\nu = 3/4$ emerges for $D < -75$ mV/nm (marked by the black arrow). (b) Measured I - V_{Probe} curve (green) and its corresponding dI/dV_{Probe} curve (blue) at fixed $D \approx -101.6$ mV/nm (white dashed line in (a)); the two valleys at filling $\nu = 1$ and $\nu = 3/4$ represent the integer and fractional insulating states.

difference sets the displacement field D , enabling combined control of doping and tunneling bias. This process provides a direct, high-resolution measurement of the local differential conductance (dI/dV_{Probe}).

Continuous and independent control over both the filling factor ν and the vertical electric displacement field D inside the tMoS₂ channel is achieved through dual-gate electrostatics. By solving the electrostatics of the probe-FLG/barrier-*h*-BN/tMoS₂/bottom-FLG junction, the net carrier density (proportional to ν) and D are governed by linear combinations of V_{Probe} and V_{PG} [see Supplementary Section S3]. Specifically, trajectories along $V_{\text{PG}} = -kV_{\text{Probe}} + \text{constant}$ (where k is the slope of the line parallel to the conduction-band edge or the $\nu = 1$ state) preserve a constant net charge (and thus constant ν), whereas trajectories along $V_{\text{PG}} = kV_{\text{Probe}} + \text{constant}$ maintain a constant displacement field D . This electrostatic transformation allows us to convert raw $dI/dV_{\text{Probe}}(V_{\text{Probe}}, V_{\text{PG}})$ transport signals directly into intuitive $dI/dV_{\text{Probe}}(\nu, D)$ phase diagrams.

Integer and fractional correlated insulating states.— Fig. 3(a) presents the measured local differential conductance dI/dV_{Probe} spectrum as a function of filling factor ν and electric displacement field D at zero magnetic field and $T = 2$ K. Two prominent dI/dV_{Probe} minima, a signature of an insulating state, emerge within the conduction band at $\nu = 1$ and $\nu = 3/4$, respectively. The integer insulating state ($\nu = 1$) persists across the entire measured range of D , corresponding to single-electron filling per moiré unit cell, consistent with the previously reported robust Mott insulating state [20]. In contrast, a fractional insulating state at $\nu = 3/4$ prominently develops as D is tuned below -75 mV/nm, as indicated by the

black arrow in Fig. 3(a). To further characterize the insulating behavior, the measured I - V_{Probe} curve (green) and corresponding dI/dV_{Probe} - V_{Probe} curve (blue) are plotted in Fig. 3(b) at $D = -101.6$ mV/nm (corresponding to the white dashed line in Fig. 3(a)), where two conductance minima, corresponding to $\nu = 1$ and $\nu = 3/4$, are clearly resolved, corresponding to the same insulating states identified in Fig. 3(a).

Fig. 4(a) displays the temperature-dependent dI/dV_{Probe} signal versus filling factor ν for temperatures ranging from 2 K to 100 K (with 2 K increments below 40 K) at a fixed displacement field $D = -101.6$ mV/nm and zero magnetic field. The insulating state at $\nu = 1$ remains robust up to 100 K, whereas the fractional insulating state at $\nu = 3/4$ vanishes above 12 K (highlighted by the bold curve). Fig. 4(b) plots the thermal degradation of dI/dV_{Probe} at $\nu = 0.75$ (blue) and $\nu = 0.81$ (red). For $\nu = 0.81$, the conductance decreases monotonically with increasing temperature, indicating metallic behavior. Conversely, at $\nu = 0.75$, the conductance initially increases with temperature before declining above 12 K, a signature characteristic of a metal-insulator transition. Generalized Wigner crystal states, emerging from strong inter-electronic correlations within the moiré flat bands, are notoriously sensitive to thermal fluctuations [2–6, 9–11, 30, 32–34], typically destabilizing above 15 K. Thus, these temperature dependencies support assigning the observed features to a Mott insulating state and a generalized Wigner crystal state in the $\approx 57^\circ$ tMoS₂.

In addition, the correlated insulating states at $\nu = 1$ and $\nu = 3/4$ identified in the $\approx 57^\circ$ device are reproducibly observed in a second QWRTD device featuring a

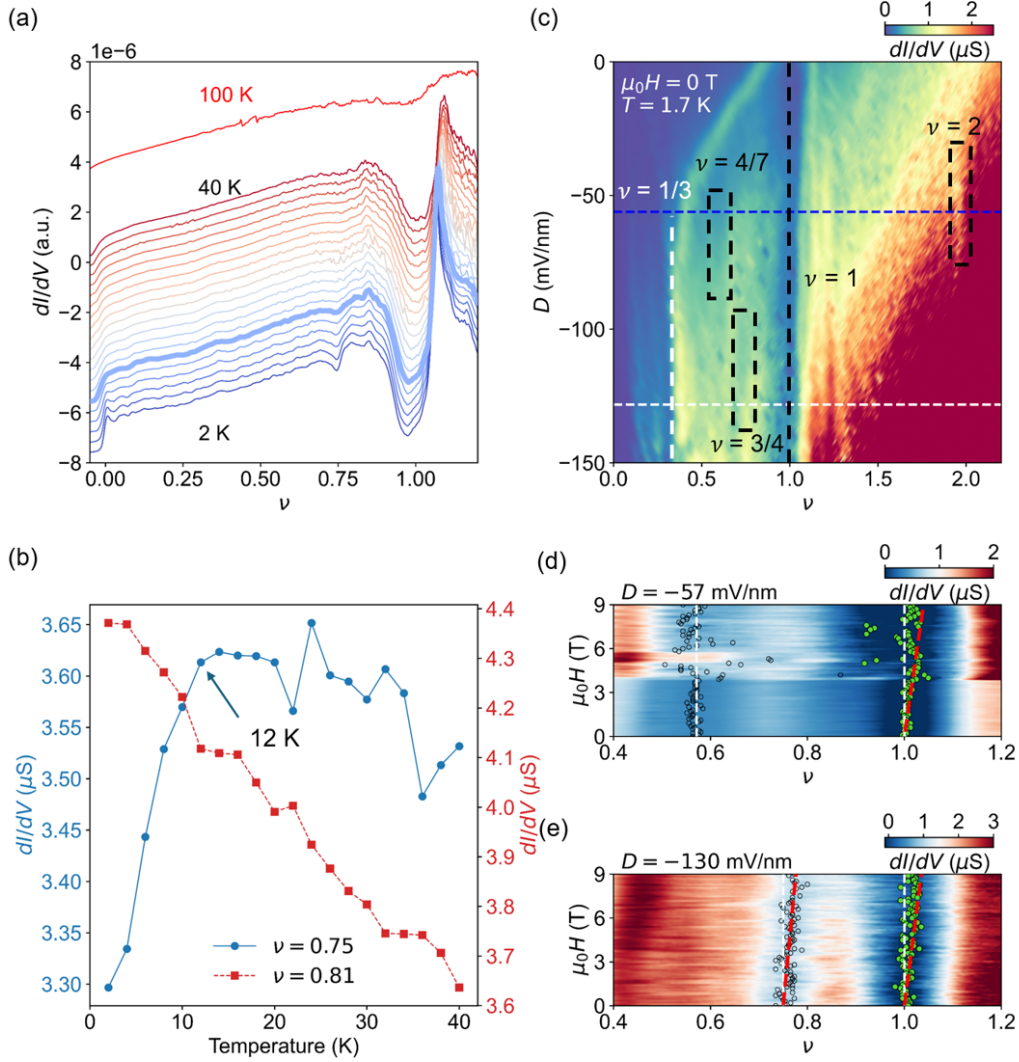


FIG. 4. (a) dI/dV_{Probe} signal versus filling factor ν as a function of temperature (2–100 K) at $D = -101.6$ mV/nm. Curves are vertically offset for clarity; the curve at 12 K, above which the $\nu = 3/4$ state disappears, is highlighted in bold. (b) Temperature dependence of the dI/dV_{Probe} signal at $\nu = 0.75$ (blue) and $\nu = 0.81$ (red). (c) dI/dV_{Probe} vs. ν and D for a second $\approx 56.5^\circ$ tMoS₂ device at 1.7 K and 0 T. Vertical dashed lines/rectangles highlight states at $\nu = 1/3, 4/7, 3/4, 1$, and 2. (d-e) dI/dV_{Probe} vs. ν and magnetic field at $D = -57$ mV/nm (horizontal blue dashed line in (c)) and $D = -130$ mV/nm (horizontal white dashed line in (c)), respectively. The red dashed lines represent the simulated Streda formula, and the white dashed lines represent the behavior of trivial states.

$\approx 56.5^\circ$ tMoS₂ moiré system (Fig. 4(c)). At smaller $|D|$, additional features emerge alongside these states, including two new insulating states at $\nu = 4/7$ and $\nu = 2$, along with a local conductance enhancement around $\nu = 1/3$; however, these additional states are no longer resolved at larger displacement fields, where only the $\nu = 3/4$ state persists, consistent with the behavior in the $\approx 57^\circ$ device. This D -induced $\nu = 3/4$ state provides direct spectroscopic evidence of strong electron correlations in near AB-stacked tMoS₂, signifying a fractional occupation of the lowest conduction miniband.

Figs. 4(d) and (e) map the differential conductance dI/dV_{Probe} as a function of filling factor ν and external

magnetic field for displacement fields $D = -57$ mV/nm and $D = -130$ mV/nm, respectively. At $D = -57$ mV/nm, two insulating states at $\nu = 4/7$ and $\nu = 1$ are observed. Upon tuning the displacement field to $D = -130$ mV/nm, the insulating state at $\nu = 4/7$ disappears, and a fractional insulating state at $\nu = 3/4$ emerges alongside the features at $\nu = 1$, consistent with the field-tuned phase transition in the $\approx 57^\circ$ device. Simulated trajectories derived from the Streda formula are indicated by red dashed lines, whereas white dashed lines denote trivial states with zero slope ($d\nu/dB = 0$). The $\nu = 1$ insulating state has a nonzero $d\nu/dB$ up to 9 T but deviates from both the predicted Streda dispersion and

the trivial-state trajectory for both $D = -57$ mV/nm and $D = -130$ mV/nm. Conversely, the $\nu = 4/7$ fractional insulating state at $D = -57$ mV/nm almost follows the white dashed line (trivial state), and the $\nu = 3/4$ fractional insulating state at $D = -130$ mV/nm appears to follow the Streda dispersion. This distinct dispersive evolution with magnetic field for the $\nu = 1$ insulating state may arise from the FLG used as both the probe and contact for the twisted MoS₂ bilayer, in which the magnetoresistance of the FLG may shift these states to a higher bias during measurement.

The interplay between flat bands and long-range Coulomb interactions gives rise to a rich phase diagram of correlated states. In moiré superlattices, the single-particle bandwidth is strongly suppressed, resulting in nearly flat bands in which the kinetic energy is quenched. At commensurate fractional fillings $\nu = p/q$ (with p and q coprime integers), a translationally invariant charge arrangement on the original moiré lattice is incompatible with the fractional filling; instead, electrons must redistribute over a superlattice with period q times that of the moiré lattice, spontaneously breaking translational symmetry to minimize the Coulomb repulsion. This spatially ordered charge pattern, which maximizes the distance between occupied sites, is known as a generalized Wigner crystal (GWC).

Specifically, at $\nu = 3/4$, there are on average 0.75 carriers per moiré unit cell, corresponding to three occupied and one vacant lattice site per four moiré cells in real space. Such charge density wave order opens a charge gap, rendering the system incompressible. In transport experiments, incompressibility manifests as a sharp peak in the longitudinal resistance, since adding or removing a carrier costs a finite charge excitation energy. Notably, the $\nu = 3/4$ GWC resides on the electron-doped side relative to half-filling ($\nu = 1/2$), and its symmetry-broken ground state — potentially including magnetic ordering — could differ qualitatively from that on the hole-doped side, providing a unique platform to explore electron-hole asymmetry and the role of quantum fluctuations in Wigner crystallization.

Conclusion.—In summary, we have demonstrated a gate-tunable QWRD as a highly sensitive, non-invasive local probe for investigating strongly correlated physics in near-AB-stacked twisted bilayer MoS₂ (tMoS₂). By converting dI/dV_{Probe} transport maps into continuous filling factor ν and displacement field D phase diagrams, we observed, in an $\approx 57^\circ$ device, a robust integer Mott insulating state at $\nu = 1$ persisting across all measured D , alongside an emergent displacement-field-induced fractional insulating state at $\nu = 3/4$ ($D < -75$ mV/nm). Temperature- and magnetic-field-dependent measurements reveal that the $\nu = 3/4$ state behaves as a generalized Wigner crystal driven by long-range Coulomb interactions, whereas the $\nu = 1$ state forms a resilient Mott insulator. These correlated states

are reproducibly observed in a second, independently fabricated device with an $\approx 56.5^\circ$ twist angle, corroborating the robustness of the underlying physics, although the precise field range over which the $\nu = 3/4$ (and $\nu = 4/7$) states emerge and vanish is less well resolved in this device. Our work extends the observation of fractional correlated electronic states to the conduction band of TMD moiré superlattices and establishes the QWRD architecture as a versatile, localized spectroscopic platform for exploring flat-band dynamics and topological phenomena in van der Waals heterostructures.

This work was supported by the Research Grants Council of the Hong Kong SAR under Grant Nos. 16302023, C6053-23G, and AoE/P-604/25R. We acknowledge the technical support from the HKUST Nano Fabrication Facility and Materials Characterization and Preparation Facility.

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- [1] R. Bistritzer and A. H. MacDonald, *Proceedings of the National Academy of Sciences* **108**, 12233 (2011).
- [2] Y. Cao, V. Fatemi, S. Fang, K. Watanabe, T. Taniguchi, E. Kaxiras, and P. Jarillo-Herrero, *Nature* **556**, 43 (2018).
- [3] Y. Xu, S. Liu, D. A. Rhodes, K. Watanabe, T. Taniguchi, J. Hone, V. Elser, K. F. Mak, and J. Shan, *Nature* **587**, 214 (2020).
- [4] Y. Zeng, Z. Xia, K. Kang, J. Zhu, P. Knüppel, C. Vaswani, K. Watanabe, T. Taniguchi, K. F. Mak, and J. Shan, *Nature* **622**, 69 (2023).
- [5] J. Cai, E. Anderson, C. Wang, X. Zhang, X. Liu, W. Holtzmann, Y. Zhang, F. Fan, T. Taniguchi, K. Watanabe, Y. Ran, T. Cao, L. Fu, D. Xiao, W. Yao, and X. Xu, *Nature* **622**, 63 (2023).
- [6] H. Park, J. Cai, E. Anderson, Y. Zhang, J. Zhu, X. Liu, C. Wang, W. Holtzmann, C. Hu, Z. Liu, T. Taniguchi, K. Watanabe, J.-h. Chu, T. Cao, L. Fu, W. Yao, C.-Z. Chang, D. Cobden, D. Xiao, and X. Xu, *Nature*, 1 (2023).
- [7] F. Wu, T. Lovorn, E. Tutuc, and A. MacDonald, *Physical Review Letters* **121**, 026402 (2018).
- [8] M. H. Naik and M. Jain, *Physical Review Letters* **121**, 266401 (2018).
- [9] E. C. Regan, D. Wang, C. Jin, M. I. Bakti Utama, B. Gao, X. Wei, S. Zhao, W. Zhao, Z. Zhang, K. Yumigeta, M. Blei, J. D. Carlström, K. Watanabe, T. Taniguchi, S. Tongay, M. Crommie, A. Zettl, and F. Wang, *Nature* **579**, 359 (2020).
- [10] H. Li, S. Li, E. C. Regan, D. Wang, W. Zhao, S. Kahn, K. Yumigeta, M. Blei, T. Taniguchi, K. Watanabe, S. Tongay, A. Zettl, M. F. Crommie, and F. Wang, *Nature* **597**, 650 (2021).
- [11] H. Li, Z. Xiang, E. Regan, W. Zhao, R. Sailus, R. Banerjee, T. Taniguchi, K. Watanabe, S. Tongay, A. Zettl, M. F. Crommie, and F. Wang, *Nature Nanotechnology* **19**, 618 (2024).
- [12] H. Li, Z. Xiang, A. P. Reddy, T. Devakul, R. Sailus,

- R. Banerjee, T. Taniguchi, K. Watanabe, S. Tongay, A. Zettl, L. Fu, M. F. Crommie, and F. Wang, *Science* **385**, 86 (2024).
- [13] J. Cai, E. Anderson, C. Wang, X. Zhang, X. Liu, W. Holtzmann, Y. Zhang, F. Fan, T. Taniguchi, K. Watanabe, Y. Ran, T. Cao, L. Fu, D. Xiao, W. Yao, and X. Xu, *Nature*, 1 (2023).
- [14] Y. Zeng, Z. Xia, K. Kang, J. Zhu, P. Knüppel, C. Vaswani, K. Watanabe, T. Taniguchi, K. F. Mak, and J. Shan, *Nature*, 1 (2023).
- [15] F. Xu, Z. Sun, T. Jia, C. Liu, C. Xu, C. Li, Y. Gu, K. Watanabe, T. Taniguchi, B. Tong, J. Jia, Z. Shi, S. Jiang, Y. Zhang, X. Liu, and T. Li, *Physical Review X* **13**, 031037 (2023).
- [16] B. A. Foutty, C. R. Kometter, T. Devakul, A. P. Reddy, K. Watanabe, T. Taniguchi, L. Fu, and B. E. Feldman, *Science* **384**, 343 (2024).
- [17] K. Kang, B. Shen, Y. Qiu, Y. Zeng, Z. Xia, K. Watanabe, T. Taniguchi, J. Shan, and K. F. Mak, *Nature* **628**, 522 (2024).
- [18] Y. Xia, Z. Han, K. Watanabe, T. Taniguchi, J. Shan, and K. F. Mak, *Nature* **637**, 833 (2025).
- [19] Y. Guo, J. Pack, J. Swann, L. Holtzman, M. Cothrine, K. Watanabe, T. Taniguchi, D. G. Mandrus, K. Barmak, J. Hone, A. J. Millis, A. Pasupathy, and C. R. Dean, *Nature* **637**, 839 (2025).
- [20] F. Wu, Q. Xu, Q. Wang, Y. Chu, L. Li, J. Tang, J. Liu, J. Tian, Y. Ji, L. Liu, Y. Yuan, Z. Huang, J. Zhao, X. Zan, K. Watanabe, T. Taniguchi, D. Shi, G. Gu, Y. Xu, L. Xian, W. Yang, L. Du, and G. Zhang, *Physical Review Letters* **131**, 256201 (2023).
- [21] Y. Wang, J. C. Kim, R. J. Wu, J. Martinez, X. Song, J. Yang, F. Zhao, A. Mkhoyan, H. Y. Jeong, and M. Chhowalla, *Nature* **568**, 70 (2019), number: 7750.
- [22] Y. Wang, J. C. Kim, Y. Li, K. Y. Ma, S. Hong, M. Kim, H. S. Shin, H. Y. Jeong, and M. Chhowalla, *Nature* **610**, 61 (2022), number: 7930.
- [23] Y. Wang, S. Sarkar, H. Yan, and M. Chhowalla, *Nature Electronics*, 1 (2024).
- [24] J. Pack, Y. Guo, Z. Liu, B. S. Jessen, L. Holtzman, S. Liu, M. Cothrine, K. Watanabe, T. Taniguchi, D. G. Mandrus, K. Barmak, J. Hone, and C. R. Dean, *Nature Nanotechnology* **19**, 948 (2024).
- [25] A. Uri, S. C. de la Barrera, M. T. Randeria, D. Rodan-Legrain, T. Devakul, P. J. D. Crowley, N. Paul, K. Watanabe, T. Taniguchi, R. Lifshitz, L. Fu, R. C. Ashoori, and P. Jarillo-Herrero, *Nature* **620**, 762 (2023).
- [26] H. Yoo, R. Engelke, S. Carr, S. Fang, K. Zhang, P. Cazeaux, S. H. Sung, R. Hovden, A. W. Tsen, T. Taniguchi, K. Watanabe, G.-C. Yi, M. Kim, M. Lusk, E. B. Tadmor, E. Kaxiras, and P. Kim, *Nature Materials* **18**, 448 (2019).
- [27] D. Wong, K. P. Nuckolls, M. Oh, B. Lian, Y. Xie, S. Jeon, K. Watanabe, T. Taniguchi, B. A. Bernevig, and A. Yazdani, *Nature* **582**, 198 (2020).
- [28] N. P. Kazmierczak, M. Van Winkle, C. Ophus, K. C. Bustillo, S. Carr, H. G. Brown, J. Ciston, T. Taniguchi, K. Watanabe, and D. K. Bediako, *Nature Materials* **20**, 956 (2021).
- [29] D. Wong, K. P. Nuckolls, M. Oh, B. Lian, Y. Xie, S. Jeon, K. Watanabe, T. Taniguchi, B. A. Bernevig, and A. Yazdani, *Nature* **582**, 198 (2020).
- [30] Y. Xie, A. T. Pierce, J. M. Park, D. E. Parker, E. Khalaf, P. Ledwith, Y. Cao, S. H. Lee, S. Chen, P. R. Forrester, K. Watanabe, T. Taniguchi, A. Vishwanath, P. Jarillo-Herrero, and A. Yacoby, *Nature* **600**, 439 (2021).
- [31] B. A. Foutty, C. R. Kometter, T. Devakul, A. P. Reddy, K. Watanabe, T. Taniguchi, L. Fu, and B. E. Feldman, "Dataset for 'Mapping twist-tuned multi-band topology in bilayer WSe₂'," (2024).
- [32] E. Wigner, *Physical Review* **46**, 1002 (1934).
- [33] J. M. Park, Y. Cao, L. Xia, S. Sun, K. Watanabe, T. Taniguchi, and P. Jarillo-Herrero, "Magic-Angle Multilayer Graphene: A Robust Family of Moiré Superconductors," (2021), arXiv:2112.10760 [cond-mat].
- [34] A. T. Pierce, Y. Xie, J. M. Park, Z. Cai, K. Watanabe, T. Taniguchi, P. Jarillo-Herrero, and A. Yacoby, *Nature Physics*, 1 (2025).